LEVELHULL · REGULATORY AND STANDARDS REGISTER

LevelHull regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/07-regulatory.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade); sources checked 2026-10-25

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The rule most likely to apply is national small vessel safety law where the kit is fitted: in Kenya, the Merchant Shipping (Small Vessel Safety) Regulations of 2012 already ask some small vessels for buoyancy that keeps them afloat when swamped, and ask that modifications to approved plans be approved, so whether fitting LevelHull needs the maritime authority's approval is the first item **to assess** with the partner. The US level flotation rule and ISO 12217-1 are design benchmarks, not obligations, for these canoes. LevelHull is not lifesaving equipment and makes no such claim. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

> **Safety:** Nothing in this register makes LevelHull a substitute for life jackets. Life jacket rules (for example the Second Schedule of the Kenyan regulations) apply to the crew whether or not the kit is fitted.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Merchant Shipping (Small Vessel Safety) Regulations, Legal Notice 114 of 2012, under the Merchant Shipping Act	Kenya	Construction, equipment and safety of commercial small vessels; First Schedule item 1 asks for built-in buoyancy able to keep the vessel afloat "when fully flooded, swamped or capsized"; regulation 37(3)(a) asks some smaller vessels for "sufficient buoyancy to keep the vessel afloat when completely swamped"; regulation 4(2) asks that modifications to approved plans be approved	Fishing canoes at a Kenyan landing are likely commercial small vessels. LevelHull is the kind of buoyancy these clauses describe; whether a fitted kit counts toward them, and whether fitting needs approval, is not known	Partner liaison with the Kenya Maritime Authority	To assess
G2	Small vessel safety rules of Uganda and Tanzania on Lake Victoria, and of other countries where the kit may be fitted	Uganda, Tanzania, others	Equivalent national rules	Same questions as G1	Partner liaison	To assess
G3	FAO, ILO and IMO, *Safety Recommendations for Decked Fishing Vessels of Less than 12 metres in Length and Undecked Fishing Vessels* (2012), including Annex 13, "Guidance on a practical buoyancy test"	International (voluntary)	Recommendations for new undecked fishing vessels, including a practical buoyancy test	Guidance for the swamp test method at TRL 4 and the case to fisheries departments; not binding	Design lead	Applies (as guidance)
G4	33 CFR 183, Subpart H and the level flotation test (183.220 to 183.235)	United States	Flotation of outboard boats under 20 ft (6.1 m); the boat is swamped "allowing water to flow between the inside and outside of the boat" and must float within 10 deg of heel	The 8 m canoes are outside its scope and outside the US; used as the design benchmark for "swamped" and "level" (PR-001 to PR-003)	Design lead	Does not apply (benchmark only)
G5	ISO 12217-1:2022, *Small craft: Stability and buoyancy assessment and*	International	Stability and buoyancy of boats 6 to 24 m, flotation of boats that can be	Applies to recreational craft placed on markets that call it up; not	Design lead	Does not apply (benchmark, to assess its

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	categorization, Part 1: Non-sailing boats of hull length greater than or equal to 6 m		swamped, design categories A to D	to these fishing canoes. A useful benchmark for flotation reserve and swamped stability		flotation clauses at TRL 4)
G6	Recreational Craft Directive 2013/53/EU	European Union	Recreational craft placed on the EU market	Not a recreational craft, not placed on the EU market	Design lead	Does not apply
G7	Personal flotation device standards (ISO 12402 series)	International	Life jackets and buoyancy aids	LevelHull is not a personal flotation device and must never be presented as one (PR-015)	Design lead	Does not apply
G8	EN 12087, *Thermal insulating products: Determination of long term water absorption by immersion* (as quoted on the foam data sheet), ISO 845 (apparent density of cellular plastics)	International	Test methods for foam water uptake and density	Test methods for the TRL 4 immersion test of PR-006 and the supplier check	Design lead	Applies (test method, TRL 4)
G9	EN 1995-1-1, Eurocode 5, and EN 338 strength classes	Europe (used as engineering practice)	Timber connections and timber strength	Used for the screw and boom checks (LVH-STM-001 S11, S13, S14, S29); not a legal requirement here	Design lead	Applies (as practice)
G10	Patent US9139267B2 (inflatable emergency float, active to about 2032)	United States	Inflatable flotation for a craft	Avoided by design: passive foam only (LVH-PRC-001)	Design lead	Met by design (not legal advice)
G11	Open hardware licences: CERN-OHL-S-2.0 (hardware), MIT (calculator), REUSE	International	Licensing	Applies to the repository	Maintainer	Met by design

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- Kenya, *The Merchant Shipping (Small Vessel Safety) Regulations*, Legal Notice 114 of 2012, Kenya Law, https://new.kenyalaw.org/akn/ke/act/ln/2012/114/eng@2022-12-31, accessed 2026-10-25.

- FAO, ILO and IMO, *Safety Recommendations for Decked Fishing Vessels of Less than 12 metres in Length and Undecked Fishing Vessels* (Rome: FAO, 2012), https://www.fao.org/4/i3108e/i3108e00.htm, accessed 2026-10-25.
- US Coast Guard, *Safe Boating: Flotation* guideline (33 CFR 183.220 to 183.235), https://www.uscgboating.org/assets/PDF/downloads/FLOTATION.pdf, accessed 2026-10-25.
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- Ravago Building Solutions, Ethafoam technical data sheet (EN 12087, ISO 845), https://ravagobuildingsolutions.com/gr/wp-content/uploads/sites/47/2022/01/tds-ethafoam-eng.pdf, accessed 2026-10-25.
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